THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{4}{5}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
2	**(b)** $\frac{4}{3}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
3	**(a)** $\frac{1}{2}$	Standard value
4	**(c)** 1	Identity
5	**(c)** 65°	$\sin\theta = \cos(90^\circ - \theta)$
6	**(b)** $\cos\theta$	$\sec\theta = 1/\cos\theta$
7	**(d)** $\frac{12}{13}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
8	**(a)** $\frac{12}{5}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
9	**(a)** $\frac{1}{2}$	Standard value
10	**(a)** 1	Identity

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\sin\theta = \frac{4}{5}$ (1) $= \frac{4}{5}$ (1).	1+1
12	$\frac{1}{2} + \frac{1}{2}$ (1) $= 1$ (1).	1+1
13	Using $\sin^2\theta + \cos^2\theta = 1$ (1); value $= 1$ (1).	1+1
14	$\sin\theta = \frac{12}{13}$ (1) $= \frac{12}{13}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AC = \sqrt{3^2+4^2} = 5$ cm (1); $\sin A = \frac{BC}{AC} = \frac{4}{5}$, $\cos A = \frac{AB}{AC} = \frac{3}{5}$ (1); $\tan A = \frac{BC}{AB} = \frac{4}{3}$ (1).	1+1+1
16	Base $= \sqrt{5^2-4^2} = 3$ (1); $\cos A = \frac{3}{5}$ (1); $\tan A = \frac{4}{3}$ (1).	1+1+1
17	Use $\sin^2 30^\circ = \frac{1}{4}$, $\cos^2 60^\circ = \frac{1}{4}$ (2); value $= \frac{1}{2}$ (1).	2+1
	OR: $AC = \sqrt{5^2+12^2} = 13$ cm (1); $\sin A = \frac{BC}{AC} = \frac{12}{13}$, $\cos A = \frac{AB}{AC} = \frac{5}{13}$ (1); $\tan A = \frac{BC}{AB} = \frac{12}{5}$ (1).	1+1+1
18	Base $= \sqrt{13^2-12^2} = 5$ (1); $\cos A = \frac{5}{13}$ (1); $\tan A = \frac{12}{5}$ (1).	1+1+1
	OR: Use $\tan^2 45^\circ = 1$, $\sin^2 90^\circ = 1$ (2); value $= 2$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypotenuse $= \sqrt{4^2+3^2} = 5$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1).	1+1+1
	OR: $PR = 10$ cm (1); $\sin P = \frac{4}{5}$, $\cos P = \frac{3}{5}$, $\tan P = \frac{4}{3}$ (2); $\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = 1 = 1$ (2).	1+2+2
20	Hypotenuse $= \sqrt{12^2+5^2} = 13$ (1); $\sin\theta = \frac{12}{13}$, $\cos\theta = \frac{5}{13}$ (1); $\sec\theta = \frac{13}{5}$ (1).	1+1+1
	OR: $PR = 39$ cm (1); $\sin P = \frac{12}{13}$, $\cos P = \frac{5}{13}$, $\tan P = \frac{12}{5}$ (2); $\left(\frac{12}{13}\right)^2 + \left(\frac{5}{13}\right)^2 = 1 = 1$ (2).	1+2+2

Mistakes to avoid:

- Treating $\sin A$ as $\sin \times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A+B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \frac{\sqrt{3}}{2}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \text{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 50°	$\sin\theta = \cos(90^\circ - \theta)$
2	**(a)** $\frac{5}{3}$	Reciprocal
3	**(c)** $\frac{15}{17}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
4	**(d)** $\frac{15}{8}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
5	**(d)** 1	Standard value
6	**(c)** 1	Identity
7	**(d)** 35°	$\sin\theta = \cos(90^\circ - \theta)$
8	**(c)** $\frac{13}{5}$	Reciprocal
9	**(c)** $\frac{24}{25}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
10	**(d)** $\frac{24}{7}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1 + 1$ (1) $= 2$ (1).	1+1
12	Using $\sin^2\theta + \cos^2\theta = 1$ (1); value $= 1$ (1).	1+1
13	$\sin\theta = \frac{15}{17}$ (1) $= \frac{15}{17}$ (1).	1+1
14	$1 + \frac{1}{2}$ (1) $= \frac{3}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AC = \sqrt{8^2 + 15^2} = 17$ cm (1); $\sin A = \frac{BC}{AC} = \frac{15}{17}$, $\cos A = \frac{AB}{AC} = \frac{8}{17}$ (1); $\tan A = \frac{BC}{AB} = \frac{15}{8}$ (1).	1+1+1
16	Base $= \sqrt{17^2 - 15^2} = 8$ (1); $\cos A = \frac{8}{17}$ (1); $\tan A = \frac{15}{8}$ (1).	1+1+1
17	Use $\cos^2 0^\circ = 1,\ \sin^2 0^\circ = 0$ (2); value $= 1$ (1).	2+1
	OR: $AC = \sqrt{7^2 + 24^2} = 25$ cm (1); $\sin A = \frac{BC}{AC} = \frac{24}{25}$, $\cos A = \frac{AB}{AC} = \frac{7}{25}$ (1); $\tan A = \frac{BC}{AB} = \frac{24}{7}$ (1).	1+1+1
18	Base $= \sqrt{25^2 - 24^2} = 7$ (1); $\cos A = \frac{7}{25}$ (1); $\tan A = \frac{24}{7}$ (1).	1+1+1
	OR: Use $\sin^2 30^\circ = \frac{1}{4},\ \cos^2 90^\circ = 0$ (2); value $= \frac{1}{2}$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypotenuse $= \sqrt{15^2 + 8^2} = 17$ (1); $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$ (1); $\sec\theta = \frac{17}{8}$ (1).	1+1+1
	OR: $PR = 34$ cm (1); $\sin P = \frac{15}{17}$, $\cos P = \frac{8}{17}$, $\tan P = \frac{15}{8}$ (2); $\left(\frac{15}{17}\right)^2 + \left(\frac{8}{17}\right)^2 = 1 = 1$ (2).	1+2+2
20	Hypotenuse $= \sqrt{24^2 + 7^2} = 25$ (1); $\sin\theta = \frac{24}{25}$, $\cos\theta = \frac{7}{25}$ (1); $\sec\theta = \frac{25}{7}$ (1).	1+1+1
	OR: $PR = 25$ cm (1); $\sin P = \frac{24}{25}$, $\cos P = \frac{7}{25}$, $\tan P = \frac{24}{7}$ (2); $\left(\frac{24}{25}\right)^2 + \left(\frac{7}{25}\right)^2 = 1 = 1$ (2).	1+2+2

Mistakes to avoid:

- Treating $\sin A$ as $\sin \times A$ and cancelling; $\sin A$ is a single ratio of the angle A.
- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \frac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A + B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \frac{\sqrt{3}}{2}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \text{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1	Standard value
2	**(b)** 1	Identity
3	**(a)** 75°	$\sin\theta = \cos(90^\circ - \theta)$
4	**(a)** $\frac{17}{8}$	Reciprocal
5	**(c)** $\frac{21}{29}$	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
6	**(c)** $\frac{21}{20}$	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
7	**(a)** 1	Standard value
8	**(d)** 1	Identity
9	**(d)** 20°	$\sin\theta = \cos(90^\circ - \theta)$
10	**(d)** $\frac{25}{7}$	Reciprocal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Using $\sin^2\theta + \cos^2\theta = 1$ (1); value $= 1$ (1).	1+1
12	$\sin\theta = \frac{24}{25}$ (1) $= \frac{24}{25}$ (1).	1+1
13	$\frac{1}{2} + 1$ (1) $= \frac{3}{2}$ (1).	1+1
14	Using $\sin^2\theta + \cos^2\theta = 1$ (1); value $= 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AC = \sqrt{20^2 + 21^2} = 29$ cm (1); $\sin A = \frac{BC}{AC} = \frac{21}{29}$, $\cos A = \frac{AB}{AC} = \frac{20}{29}$ (1); $\tan A = \frac{BC}{AB} = \frac{21}{20}$ (1).	1+1+1
16	Base $= \sqrt{29^2 - 21^2} = 20$ (1); $\cos A = \frac{20}{29}$ (1); $\tan A = \frac{21}{20}$ (1).	1+1+1
17	Use $\tan^2 45^\circ = 1,\ \sin^2 30^\circ = \frac{1}{4}$ (2); value $= \frac{3}{4}$ (1).	2+1
	OR: $AC = \sqrt{9^2 + 40^2} = 41$ cm (1); $\sin A = \frac{BC}{AC} = \frac{40}{41}$, $\cos A = \frac{AB}{AC} = \frac{9}{41}$ (1); $\tan A = \frac{BC}{AB} = \frac{40}{9}$ (1).	1+1+1
18	Base $= \sqrt{41^2 - 40^2} = 9$ (1); $\cos A = \frac{9}{41}$ (1); $\tan A = \frac{40}{9}$ (1).	1+1+1
	OR: Use $\sin^2 90^\circ = 1,\ \cos^2 0^\circ = 1$ (2); value $= 2$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypotenuse $= \sqrt{21^2 + 20^2} = 29$ (1); $\sin\theta = \frac{21}{29}$, $\cos\theta = \frac{20}{29}$ (1); $\sec\theta = \frac{29}{20}$ (1).	1+1+1
	OR: $PR = 87$ cm (1); $\sin P = \frac{21}{29}$, $\cos P = \frac{20}{29}$, $\tan P = \frac{21}{20}$ (2); $\left(\frac{21}{29}\right)^2 + \left(\frac{20}{29}\right)^2 = 1 = 1$ (2).	1+2+2
20	Hypotenuse $= \sqrt{40^2 + 9^2} = 41$ (1); $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$ (1); $\sec\theta = \frac{41}{9}$ (1).	1+1+1
	OR: $PR = 82$ cm (1); $\sin P = \frac{40}{41}$, $\cos P = \frac{9}{41}$, $\tan P = \frac{40}{9}$ (2); $\left(\frac{40}{41}\right)^2 + \left(\frac{9}{41}\right)^2 = 1 = 1$ (2).	1+2+2

Mistakes to avoid:

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- Mixing up the sides: for angle A, perpendicular is the side *opposite* A and base is the side *adjacent* to it.
- Writing $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$; that is $\cot\theta$.
- Writing $\sin 2A = 2\sin A$ or $\sin(A + B) = \sin A + \sin B$, which are false.
- Wrong table values, e.g. $\sin 30^\circ = \dfrac{\sqrt{3}}{2}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$.
- Using the wrong identity: $1 + \cot^2\theta = \text{cosec}^2\theta$, not $\sec^2\theta$; and forgetting that $\tan 90^\circ$ is not defined.

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